

Using dye tracer for visualizing roots impact on soil structure and soil porous system

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Abstract: Plants influence the water regime in soil by both water uptake and an uneven distribution of water infiltration at the soil surface. The latter process is more poorly studied, but it is well known that roots modify soil structure by enhancing aggregation and biopore production. This study used a dye tracer to visualize the impact of plants on water flow in the topsoil of a Greyic Phaeozem. Brilliant blue was ponded to 10 cm height in a 1 m × 1 m frame in the field immediately after harvest of winter wheat (*Triticum aestivum* L.). After complete infiltration, the staining patterns within the vertical and horizontal field-scale sections were studied. In addition, soil thin sections were made and micromorphological images were used to study soil structure and dye distribution at the microscale. The field-scale sections clearly documented uneven dye penetration into the soil surface, which was influenced by plant presence and in some cases by mechanical compaction of the soil surface. The micromorphological images showed that root activities compress soil and increase the bulk density near the roots (which could be also result of root water uptake and consequent soil adhesion). On the other hand in few cases a preferential flow along the roots was observed.

Key words: field sections; macro-scale; micro-scale; micromorphological images; plant; ponding dye infiltration; roots; soil structure.

Introduction

Water flow and contaminant transport in soils is frequently influenced by preferential flow. Various dye tracers are frequently used to study preferential flow phenomena (Flury & Wai 2003). The most often applied dye is the food color Brilliant Blue FCF, although fluorescent tracers (Gerke et al. 2013) and other compounds may also be applied. There are many studies dealing with the Brilliant Blue FCF characteristics (Flury & Flühler 1995; Morris et al. 2008) and studies using this dye for visualization of non-equilibrium water flow and dye transport in soils (among others Flury et al. 1994; Morris & Mooney 2004; Kasteel et al. 2005, 2007; Person 2005; Sander & Gerke 2007, 2009; Bogner et al. 2008; Mooney & Morris 2008; Anderson et al. 2009; Bachmair et al. 2009; Cey & Rudolph 2009; Homolák et al. 2009; Nobles et al. 2010; Hardie et al. 2011; Lichner et al. 2011, 2012, 2013a,b; Kodešová et al. 2012; Etana et al. 2013). Previous studies showed that the dye tracer distribution (indicating preferential flow variation in different soil types), may be significantly enhanced by earth-worm burrows (Cey and Rudolph 2009; Sander and Gerke 2009; Kodešová et al. 2012), root channels (Nobles et al. 2010; Kodešová et al. 2012) or straw incorporation (Kasteel et al. 2007; Kodešová

et al. 2012), depends on land use (Bachmair et al. 2009; Homolák et al. 2009) and soil surface topology (Lichner et al. 2011), occurrence of surface cracks (Sander & Gerke 2007) or subsurface fractures (Kodešová et al. 2012), existence of subsurface compaction (Kodešová et al. 2012; Etana et al. 2013), presence of macropores between well-developed soil aggregates covered by organomineral coating (Kodešová et al. 2012), antecedent soil water content (Hardie et al. 2011) and so on.

Dye tracer transport is primarily influenced by soil structure, which the research described thus far has investigated at the macro-scale, e.g. on images of field-scale or laboratory container-scale soil sections. In these studies the smallest sections of stained soils were made from soil blocks with dimensions of 20 cm × 20 cm × 20 cm (Morris & Mooney, 2004; Mooney & Morris, 2008). There has been little attempt to study the dye pattern at the micro-scale, e.g. on images of soil sections at the micromorphological (pore-scale) level (i.e. blocks with dimensions equal or smaller than 1 cm × 1 cm × 1 cm).

Nobles et al. (2004) performed measurement of the distribution of bromide and Brilliant Blue FCF in soil at the micro-scale. They showed that the movement of the dye was significantly retarded in soil, as compared

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Table 1. Depths of soil horizons and basic soil properties determined on grab soil samples: clay, silt and sand content, and specific density (Kodešová et al. 2012).

Horizon	Depth (cm)	Clay (%)	Silt (%)	Sand (%)	Specific density (g cm^{-3})
Ap	0–25	27.6	59.2	13.2	2.58
Bth	25–44	30.3	57.9	11.8	2.57
Ck	44–125	24.0	63.9	12.1	2.61

to the mobility of the Br tracer. Further research using the both tracers by Kasteel et al. (2002) and Nobles et al. (2010), attributed the difference of Br and dye transport to dye adsorption on soil particles and the larger size of the Brilliant Blue FCF organic molecule compared to Br. Kodešová et al. (2012) presented visualization of preferential flow at both the macro- (field sections – mentioned above) and micro-scale (thin soil sections). Micromorphological images showed that, in the case of a Haplic Luvisol (where accumulated clay in the Bt horizons formed coatings on the aggregate surfaces), the dye was primarily distributed either in the interaggregate pores and then in the pores inside the aggregates, or in the isolated large pores connected to the dye source and then into the matrix pores. The dye distribution in the soil matrix was uneven as well. Accumulated organic matter, clay coatings (Fér & Kodešová 2012) and isolated larger capillary pores, which initially did not contain the dye tracer, behaved as less-permeable or impermeable barriers. It was postulated that lower dye mobility through clay and organic coatings could be caused by the size of the Brilliant Blue FCF molecule. There was also an uneven distribution of the dye, which was possibly caused by the hierarchical pore size distribution of the soil matrix. Results indicated a multi-modal character of preferential flow in this soil. Kodešová et al. (2012) also studied a Haplic Cambisol, where the clay skins on ped surfaces were not present. In this soil the dye pattern studied at the micro-scale was mostly affected by fractures, size and shape of mineral grains.

Another factor that greatly influences soil structure is plant roots. The architecture of plant roots (i.e. root distribution, length, diameter, strength etc.) depends on the plant species and its interaction with soil conditions (Bengough et al. 2011; Hallett et al. 2013; Loades et al. 2013). Although a large body of research has measured the preferential flow of water along the surfaces of live roots and through empty biopores that remain following the decomposition of dead roots, less research has studied the processes at the micro-scale and most of this research relies on non-invasive imaging of extracted soil cores or thin soil plates that do not represent natural conditions (e.g. Moran et al. 2000; Oswald et al. 2008; Moradi et al. 2009, 2013; Rudolph et al. 2012; Stingaciu et al. 2013; Rudolph-Mohr et al. 2014). The studies dealing with root impact on soil structure at the micromorphological scale mostly concentrated on their and other organic litter impact on structure of decomposed organic matter (e.g. Kodešová et al. 2007; Nikodem et al. 2013). The goal of this study

was therefore to visualize preferential flow due to plant roots influences at macro- and micro-scale using similar approach as described by Kodešová et al. (2012). By studying soil *in situ*, this study provides missing information required to upscale from rhizosphere to plot scale processes. It is directly relevant to understanding water capture by plants and the potential preferential flow of fertilizers and other chemicals to the root-soil interface.

Material and methods

The study was performed at the experimental station of the Crop Research Institute in Čáslav in the Czech Republic. The conventional tillage with the 3-year rotation system was applied at selected field. The studied soil was a Greyic Phaeozem with a loess parent material. Three soil horizons Ap, Bth and Ck were identified (Table 1). Studies by Kodešová et al. (2009a,b) reported a relatively homogeneous matrix structure with many isolated pores in all 3 horizons. Stabilities of aggregates in the Ap and Bth horizons were high, probably due to the presence of clay and organic matter coatings. Different soil-porous structure of the horizons documented in micromorphological images (Kodešová et al. 2008, 2009b) resulted in considerably different hydraulic properties and strongly impacted observed herbicide transport in this soil (Kodešová et al. 2008, 2009b; Kočárek et al. 2005, 2010). Finally, Jirků et al. (2013) documented temporal (seasonal and annual from 2007 to 2010) variability of soil structure and hydraulic properties of topsoil.

The field dye infiltration experiment was carried out in August 2010 [immediately after harvest of winter wheat (*T. aestivum*)] using a similar procedure as Sander & Gerke (2007) and Kodešová et al. (2012). An aluminum infiltration frame (area of $1 \text{ m} \times 1 \text{ m}$ and height of 30 cm) was gently inserted into the soil to a depth of 20 cm using a sharp knife and a plastic hammer. Site was selected to maximally reduce wheel truck soil compaction. Then 100 liters of solution with Brilliant Blue FCF (5 kg m^{-3}) was infiltrated into the soil from an initial ponding depth of 10 cm. The infiltration plot was covered by plastic to prevent evaporation and/or rainfall. On the next day, one half of the plot was excavated in a series of horizontal slices (at selected depths – 10 slices) and another half was excavated in a series of vertical slices (2.5 cm apart) to study the dye distribution within the soil profile to a depth of 100 cm. Photographs of the entire vertical or horizontal slides were taken using a NIKON D700 camera with a resolution of 12.1 Megapixels, and fitted with a NIKON 24–85 mm zoom lens with a minimum aperture of F2.8–4.

During the vertical and horizontal soil profile excavations, large undisturbed stained soil samples of 10 wide $\times$ 8 deep $\times$ 4 thickness cm were taken from selected horizons and depths (here we show only topsoil) to characterize various mechanisms of the dye distribution at the micro-scale.

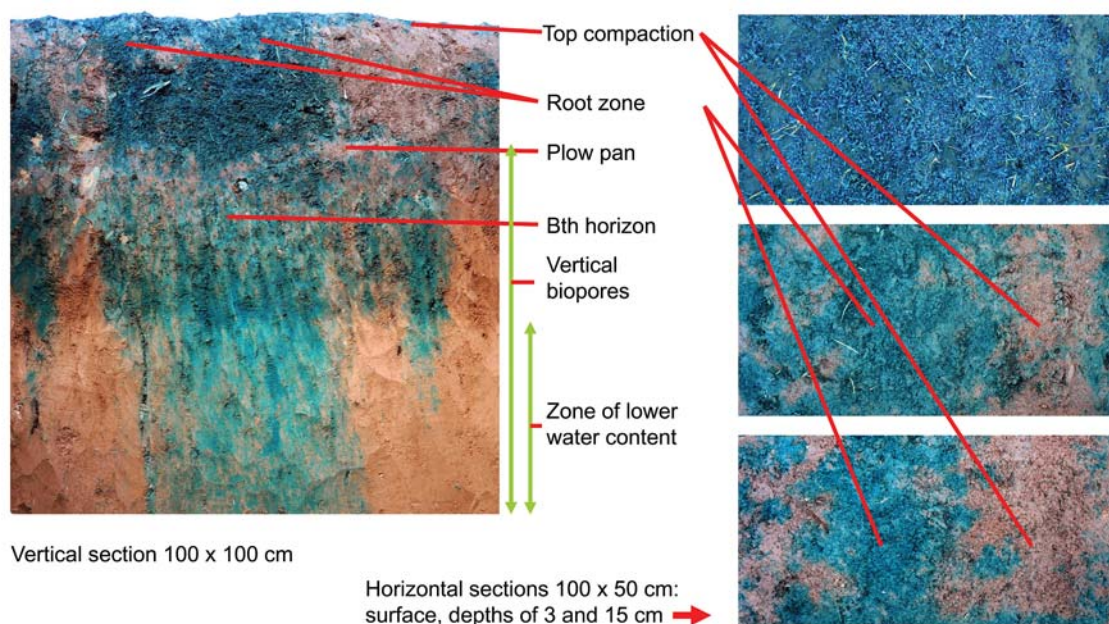


Fig. 1. Field soil sections.

Micromorphological properties characterizing the soil pore structure and the dye distribution were studied on thin soil sections prepared from the soil samples. Thin sections were prepared according to the methods presented by Stoops (2003). Before thin section preparation the larger samples were divided into 6 smaller samples. The final thin section size was approximately 1.5 cm by 2 cm. Images were taken with a polarization microscope OLYMPUS BX51 with the digital camera OLYMPUS DP70 (using software Deep Focus 3.0) at one magnification (2X) and a resolution of 300 dpi. The scanner Epson Perfection V750 PRO (image resolution of 1200 × 1200 dpi) was also used to scan divided soil blocks used to prepare thin soil sections and also thin sections.

Results and discussion

An example of dyed field sections is shown in Fig. 1. Both vertical transport down the soil profile and horizontal transport at various depths from the surface to 15 cm depth were visualized. Despite that the spot to insert aluminum frame was selected to minimize the heavy machinery impact on soil structure, it is evident that the dye pattern within the vertical section was affected by soil surface mechanical compaction (caused probably by single harvester crossing), which was not apparent at the soil surface. While a limited amount of dye tracer penetrated through the compacted soil surface, a relatively homogeneous dye distribution was observed in the Ap horizon under the uncompacted surface. Horizontal images from the depths of 2 cm and 15 cm show enhanced dye penetration along root channels and in regions with a higher fraction of larger interaggregate pores formed in the root zone. It is evident that roots even caused dye penetration through the surface influenced by mechanical compaction by providing a continuous flow path to depth.

From the dye pattern (vertical field section) there is evidence that a relatively thin plow pan (the very

compact soil layer presumably of very low permeability) had formed below the surface tilled horizon (as in Etana et al. (2013) and Kodešová et al. (2012)). Here the preferential flow occurred in large biopores (mostly earthworm burrows) and fractures. In the case of deeper Bth horizons, the preferential flow occurred due to gravitational biopores formed from the vertical burrows of earthworms. Dye tracer also revealed a prismatic structure and the presence of isolated large capillary pores (similarly to the Bt1 and Bt2 in the Haplic Luvisol studied by Kodešová et al. (2012)). Dye transport in the Ck horizon was also influenced by the gravitational biopores, but mostly by large capillary pores typical for loess (Kodešová et al. 2006, 2009b). The vertical field section also indicated that dye percolation was influenced by considerably lower soil-water content at the bottom of the soil profile. It seemed that while dye tracer penetrated from the macropores into the matrix pores in the zone of higher soil-water content, it remained in macropores in zones with considerably lower soil-water content. This may be attributed to dye diffusion into matrix pores, which could not happen in the dry zone.

Figure 2 illustrates the formation of soil thin sections from impregnated blocks to cut and polished 30 µm slices. In this example the soil block was taken from a region where the soil from a macroscale point of view was uniformly stained. However, Fig. 2 shows that at the microscale dye was not evenly distributed around the plant and its roots. A number of processes could drive this uneven distribution. Roots and stems compress soil and increase soil bulk density near to the plant (Hillel 2004). The soil bulk density increase could also result from root water uptake and consequent soil adhesion (Young 1998). As a result, dye had not penetrated into the compacted regions and was distributed in less dense regions. The reason could be lower permeability of dense regions and the size of the Brilliant Blue

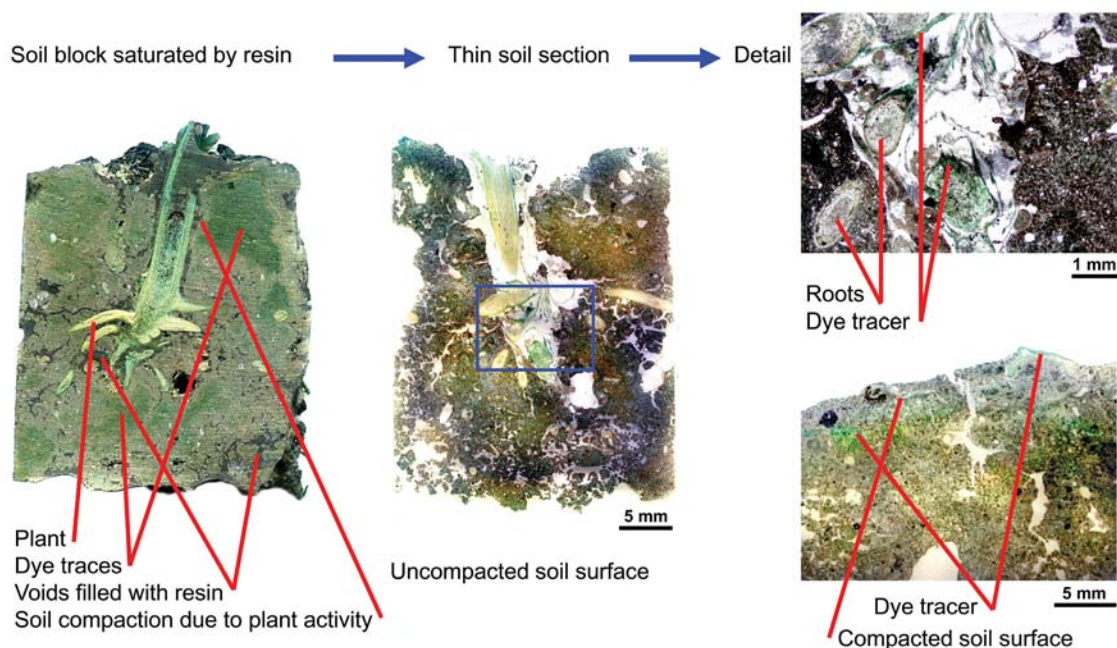


Fig. 2. Soil sample, thin soil sections and micromorphological image.

FCF organic molecule (Nobles et al. 2010; Kodešová et al. 2012).

In a few cases preferential flow along the roots was observed. It should be also noted that a large spaces between the plant and soil (visible in the soil block and thin soil section) could be created when drying soil samples. Figure 2 also shows difference between uncompacted and compacted soil surface. While dye was not evenly distributed but penetrated into the entire soil sample in the case of an uncompacted soil surface, in the case of compacted soil surface it stained only a very thin layer at the surface.

In conclusion, our results confirmed a multi-modal character of preferential flow in the top horizon. The field-scale sections clearly documented uneven dye penetration into the soil surface, which increased due to presence of roots and in some cases decreased due to mechanical compaction of the soil surface. The micromorphological images showed that roots and stalk grow compressed soil and increased the bulk density near the roots and dye trace could not penetrate into these regions. On the other hand a preferential flow along the roots was observed.

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